American Computer Science League

2022-2023 ● Contest 3: Short Problems Solutions ● Classroom Division

1. Boolean Algebra

$$(A + B) + AB + B = A + B + AB$$
$$= A + B(1 + A)$$
$$= A + B$$

D. $A + B$

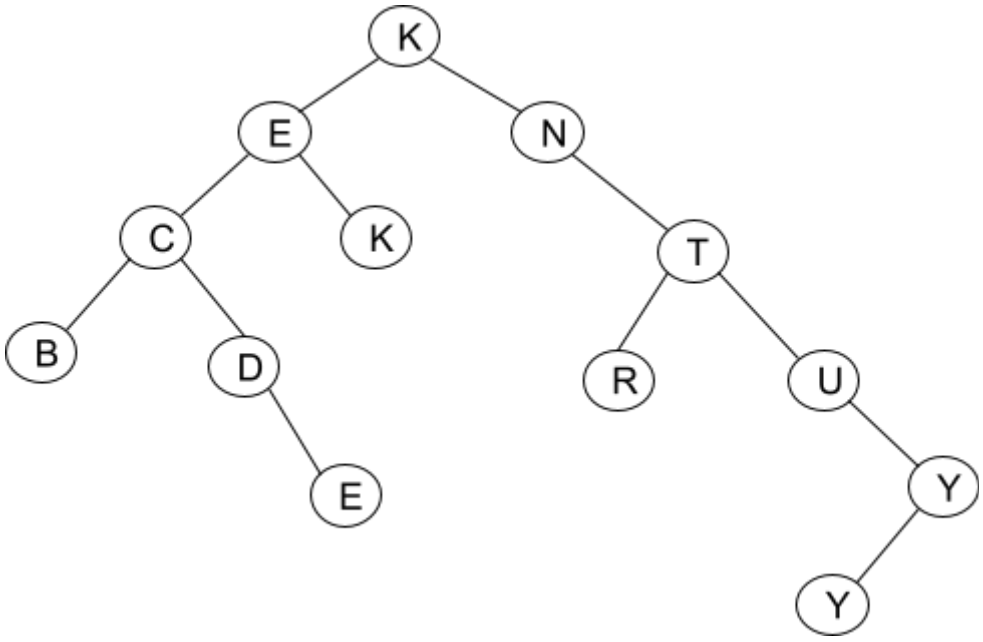
2. Boolean Algebra

$$(\overline{A}\,B)(A + \overline{B}) = \overline{A}\,BA + \overline{A}\,B\,\overline{B} = 0\,\overline{A}\,B + 0 = 0$$

A. 0

3. Data Structures

The binary search tree for K E N T U C K Y D E R B Y is:



The nodes at depth 3 from left to right are BDRU.

B. BDRU

4. Data Structures

The stack is constructed using LIFO as follows:

M, ME, MER, ME, M, MC, MCH, MC, M, MA, M, MN, MNT, MNTO, MNTOF, MNTO, MNTOV, MNTO, MNTOE, MNTOEN, MNTOE, MNTO, MNTOI, MNTO, MNTOC, MNTOCE, MNTOCE, MNTOC, MNTO

The next item popped would be: O

D. O

5. FSAs and Regular Expressions

The following do not satisfy the given regular expression.

[a-m]*[^emtsy]*t.*y

1. crusty fails at s
2. documentary fails at m
3. slapstick fails at first s

B. 3

6. Boolean Algebra

$$\begin{aligned}\overline{A}(A + \overline{B}) + \overline{A}\,B &= \overline{A}A + \overline{A}\,\overline{B} + \overline{A}\,B \\ &= \overline{A}\,\overline{B} + \overline{A}\,B \\ &= \overline{A}(\overline{B} + B) \\ &= \overline{A}\end{aligned}$$

B. $\overline{A}$

7. Boolean Algebra

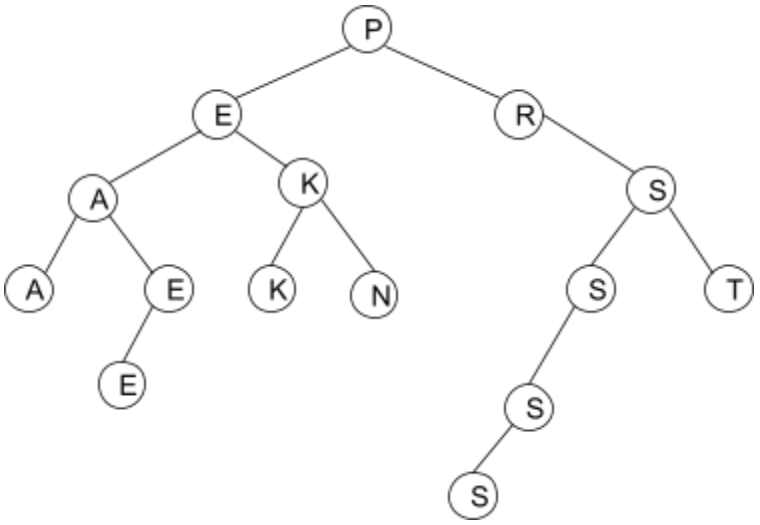
$$\begin{aligned}(\overline{A\,\overline{B} + C})\,(\overline{B\,\overline{C} + A}) &= \overline{A}\,\overline{\overline{B}}\,\overline{C}\,\overline{B}\,\overline{\overline{C}}\,\overline{A} \\ &= \overline{A}\,\overline{C}\,(\overline{A} + \overline{\overline{B}})(\overline{B} + \overline{\overline{C}}) \\ &= \overline{A}\,\overline{C}\,(\overline{A}\,\overline{B} + \overline{A}\,C + B\,\overline{B} + B\,C) \\ &= \overline{A}\,\overline{B}\,\overline{C} + \overline{A}\,C\,\overline{C} + 0 + \overline{A}\,B\,C\,\overline{C} \\ &= \overline{A}\,\overline{B}\,\overline{C}\end{aligned}$$

$\overline{A}\,\overline{B}\,\overline{C} = 1 \Rightarrow \overline{A} = 1,\ \overline{B} = 1, \overline{C} = 1$
So (0, 0, 0) is the only triple that makes the expression TRUE.

A. 1

8. Data Structures

The binary search tree for PREAKNESSSTAKES is as follows:



There are 3 nodes with only a left child: E,S, and S.

B. E, S, S

9. Data Structures

The queue is constructed using FIFO as follows:

R, RO, ROM, OM, M, ME, MEO, EO, O, OA, A, AN, AND, ANDJ, ANDJU, NDJU, NDJUL, DJUL, DJULI, DJULIE, JULIE, ULIE, ULIET, LIET

The next item popped would be L.

D. L

10. FSAs and Regular Expressions

regular expression: [b-ps-z]* [aei] [j-p]*

Dances:
- waltz: fails at t in [j-p]*
- tango: fails at g in [j-p]*
- chacha: fails at 2nd c in [j-p]*
- polka: matches
- samba: fails at b in [j-p]*
- foxtrot: fails at r in [aei]
- charleston: fails at r in [aei]
- paso: fails at s in [j-p]*

B. 1

salsa: fails at 2nd s in [j-p]* rhumba: fails at r in [b-ps-z]* mambo: fails at b in [j-p]* quickstep: fails at q in [b-ps-z]* Only 1 matches: polka	